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Systems and methods for sequential energization of heating elements in an electric catalyst unit for ammonia dissociation

Abstract

The present invention relates, in general, to a system and method for sequentially energizing heating elements in an electric catalyst unit for ammonia dissociation on-board a vehicle. The present invention utilizes output temperature readings at various sections within the electric catalyst unit, and sequentially energizes corresponding heating elements only if the output temperature of an upstream section is below a threshold temperature required for ammonia dissociation to occur. By sequentially energizing heating elements as needed, versus fully energizing every heating element, the present invention mitigates the risk of degradation and failure of the vehicle power system and other electrical components in the vehicle.

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Background/Summary

BACKGROUND

Field of the Invention

(1) The present invention relates, in general, to systems and methods for sequentially energizing heating elements as needed within an electric catalyst unit to promote ammonia dissociation.

Description of Related Art

(2) In an on-board ammonia dissociation system for vehicles, an electric catalyst unit can be utilized during a cold start of an internal combustion engine, or during low load engine operation, where the temperature of the exhaust gas from the engine is relatively low. Electric catalyst units can heat a catalyst to a temperature sufficient to perform ammonia dissociation, however, conventional electric catalyst units require a significant amount of power to do so.

(3) Commercially available catalyst units for ammonia dissociation are typically large and bulky industrial systems that use AC voltage, such as, for example, systems manufactured by Thermal Dynamix Inc.™ of Westfield, MA. These industrial ammonia dissociation systems are not suitable for use on-board vehicles given their size, weight, and large voltage requirements.

(4) Conventional electric catalyst units for vehicles, also referred to as catalytic converters, are well known. These catalyst units typically have a planar metallic conductor through which an electric current is passed. For example, Emitec Technologies GmbH™ of Lohmar, Germany manufactures an electric catalytic converter that is used to treat exhaust gas emissions, and includes a spiral planar conductor. Such planar conductors are used for exothermic reactions.

(5) The ammonia dissociation reaction is highly endothermic, meaning that a significant amount of heat is required to break the chemical bonds of ammonia molecules. The high temperature, and thus higher energy, needed to crack ammonia can result in an inefficient and expensive process to

generate hydrogen within an on-board electric catalyst unit.

(6) The energy to heat the electric catalyst unit is typically provided by a vehicle battery. For constant stop-and-go conditions, where the electric catalyst unit is utilized to initiate the hydrogen production, high current in repeatedly drawn from the vehicle battery. This continuous and repeated current draw from the vehicle battery poses the risk of rapidly draining the battery, potentially leaving the vehicle unable to start, and in extreme cases, even damaging the battery itself due to excessive heat generation from the high current flow; this is especially concerning if the high current draw persists over a prolonged period. Furthermore, when a high current is drawn, the battery voltage can drop significantly, impacting the performance of other electrical components in the vehicle.

(7) In addition, typically electric catalyst units have a single heating element that is always fully energized—either the heating element is drawing its full current or it is drawing no current. This binary operation could result in excessive or unneeded electric current being drawn from the power supply if the heating element can provide sufficient heat with a lower heater percentage.

(8) Therefore, there is a need for an electric catalyst unit capable of reaching temperatures sufficient to perform ammonia dissociation in an efficient manner on-board a vehicle, and which addresses the challenges and drawbacks of repeated high current draws from the vehicle battery.

SUMMARY

(9) In an embodiment, the present invention is directed to an electric catalyst unit for ammonia dissociation, comprising: a housing comprising a first section and a second section, the first section containing a first heating element and the second section containing a second heating element; a first power feed-through coupled to the first heating element and to a power supply; a second power feed-through coupled to the second heating element and to the power supply; a first temperature sensor coupled to the first section; a second temperature second coupled to the second section; and a controller communicatively coupled to power supply, the first temperature sensor, and the second temperature sensor, wherein the controller increases a heater percentage of the first heating element if a temperature detected by the first temperature sensor is below a threshold temperature, and wherein the controller supplies power to the second heating element only if (i) the first heating element is operating at a maximum heater percentage and (ii) the temperature detected by the first temperature sensor is below the threshold temperature.

(10) In another embodiment, the present invention is directed to an electric catalyst unit for ammonia dissociation, comprising: a housing comprising a first section and a second section, the first section containing a first heating element and the second section containing a second heating element; a first power feed-through coupled to the first heating element and to a power supply; a second power feed-through coupled to the second heating element and to the power supply; a first temperature sensor mounted to the first section downstream from the first heating element; a second temperature second mounted to the second section downstream from the second heating element; and a controller communicatively coupled to power supply, the first temperature sensor, and the second temperature sensor, wherein the controller increases a heater percentage of the first heating element if a temperature detected by the first temperature sensor is below a threshold temperature, and wherein the controller supplies power to the second heating element only if (i) the first heating element is operating at a maximum heater percentage and (ii) the temperature detected by the first temperature sensor is below the threshold temperature.

(11) In yet another embodiment, the present invention is directed to an electric catalyst unit for ammonia dissociation, comprising: a housing comprising a first section and a second section, a first heating element disposed within the first section, the first heating element having a positive end coupled to a first power feed-through and having a negative end coupled to a first ground terminal; a second heating element disposed within the second section, the second heating element having a positive end coupled to a second power feed-through and having a negative end coupled to a second ground terminal; a first temperature sensor mounted to the first section at a location

downstream from the first heating element; a second temperature second mounted to the second section at a location downstream from the second heating element; and a controller communicatively coupled to a power supply, the first temperature sensor, and the second temperature sensor, wherein the controller increases a heater percentage of the first heating element if a temperature detected by the first temperature sensor is below a threshold temperature, and wherein the controller supplies power to the second heating element only if (i) the first heating element is operating at a maximum heater percentage and (ii) the temperature detected by the first temperature sensor is below the threshold temperature.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other embodiments of the present invention will be discussed with reference to the following exemplary and non-limiting illustrations, in which like elements are numbered similarly, and where:

(2) FIG. **1** is a perspective view of an on-board ammonia dissociation system for an internal combustion engine;

(3) FIG. **2** is a perspective view of an electric catalyst unit, according to an embodiment of the present invention;

(4) FIG. **3** is a cross-sectional side view of the electric catalyst unit, according to an embodiment of the present invention;

(5) FIG. **4** is a cross-sectional perspective view of the electric catalyst unit, according to an embodiment of the present invention;

(6) FIG. **5** is a lateral side view of the electric catalyst unit, according to an embodiment of the present invention;

(7) FIG. **6** is a perspective view of a section of the housing of the electric catalyst unit, according to an embodiment of the present invention;

(8) FIG. **7** is an inlet end view of the electric catalyst unit, according to an embodiment of the present invention;

(9) FIG. **8** is an inlet end view of the electric catalyst unit with a cover removed, according to an embodiment of the present invention;

(10) FIG. **9** is a flowchart depicting sequential energization of heating elements as gas flows through the electric catalyst unit, according to an embodiment of the present invention;

(11) FIG. **10** is a flowchart depicting, sequential energization of heating elements as gas flows through the electric catalyst unit, according to an embodiment of the present invention; and

(12) FIG. **11** is a block diagram of the control system for the electric catalyst unit, according to an embodiment of the present invention.

DEFINITIONS

(13) The following definitions are meant to aid in the description and understanding of the defined terms in the context of the present invention. The definitions are not meant to limit these terms to less than is described throughout this specification. Such definitions are meant to encompass grammatical equivalents.

(14) As used herein, the term “vehicle” refers to any moving vehicle that is capable of carrying one or more human occupants and/or cargo, or which is capable of performing a task, and which is powered by any form of energy. The term “vehicle” includes, but is not limited to: (a) motor vehicles such as cars, trucks, vans, minivans, sport utility vehicles, passenger carrying vehicles, goods carrying vehicles, 2-, 3-, and 4-wheeled vehicles, quadricycles, motorcycles, scooters, all-terrain vehicles, utility task vehicles, and the like; (b) airborne vehicles such as helicopters, airplanes, airships, drones, aerospace vehicles, and the like; (c) marine vessels such as dry cargo

ships, liquid cargo ships, specialized cargo ships, tug-boats, cruise ships, recreational boats, fishing boats, personal watercraft, jet skis, and the like; (d) locomotives; and (e) heavy equipment and machinery, power generators, lawnmowers and tractors, agricultural equipment and machinery, forestry equipment and machinery, construction equipment and machinery, mining equipment and machinery, and the like.

(15) As used herein, the term “internal combustion engine” refers to any engine, spark ignition gasoline engine, compression ignition diesel engine, rotary, reciprocating, or other engine wherein combustion takes place in a combustion chamber, such that the products of combustion, together with any other by-products, perform work by exerting force on a moving surface from which the mechanical output is obtained from the engine. The term “internal combustion engine” includes, but is not limited to, hybrid internal combustion engines, two-stroke engines, four-stroke engines, six-stroke engines, and the like.

(16) As used herein, the term “catalyst” refers to a material that promotes a chemical reaction. The term “catalyst” includes, but is not limited to, a catalyst or catalysts capable of promoting dissociation reactions, such as ammonia cracking reactions, whether used as base catalyst(s) and/or additive catalyst(s). The catalyst, for the purposes of the present invention, can include, but is not limited to, a non-stoichiometric lithium imide, nickel, iron, cobalt, iron cobalt, ruthenium, vanadium, palladium, rhodium, platinum, sodium amide, and the like, as well as various combinations thereof.

(17) As used herein, the terms “dissociation” and “cracking” refer to a process or processes by which ammonia is dissociated and/or decomposed into constituent hydrogen and nitrogen components over at least one catalyst.

(18) As used herein, the term “nickel alloy” refers to pure nickel or an alloy containing nickel as a main component. The term “nickel alloy” includes, but is not limited to, Inconel®, such as, for example, Inconel® 625, Inconel® 718, Inconel® 725, and other compound metals having nickel as a main component. Inconel® is the trademark of Special Metals Corporation of Huntington, West Virginia, and is a nickel-chromium-based superalloy often utilized in extreme environments where components are subjected to high temperature, pressure, or mechanical loads.

(19) As used herein, the term “ceramic” refers to silicon nitride ceramic, steatite, and other non-conductive ceramic materials.

(20) As used herein, the term “lattice” refers to a structure where unit cells are repeated at one or more respective points of a periodic array, resulting in a structure that appears the same from any point.

(21) As used herein, the term “three-dimensional printing” and “three-dimensionally printed” refer to a three-dimensional object obtained via an additive manufacturing process, where the object has a height, a width, and a length. Additive manufacturing processes are those in which material is deposited, joined, or solidified under computer control, with the material being added together, typically layer by layer.

(22) As used herein, the terms “seal” and “sealed” refer to protection from harmful effects of ambient environmental conditions. Such protection includes protection against differences in pressure, temperature, fluid/humidity, electrical potential, shock, and gaseous compositions. These terms also refer to a hermetic, vacuum, air-tight, and/or gas-tight environment within a housing, such as in a pressure vessel.

DETAILED DESCRIPTION

(23) It should be understood that aspects of the present invention are described herein with reference to the figures, which show illustrative embodiments. The illustrative embodiments herein are not necessarily intended to show all embodiments in accordance with the invention, but rather are used to describe a few illustrative embodiments. Thus, aspects of the invention are not intended to be construed narrowly in view of the illustrative embodiments. In addition, although the present invention is described with respect to its application for an internal combustion engine for a

vehicle, it is understood that the system could be implemented in any engine-driven setting that may be powered by ammonia and/or hydrogen fuel.
(24) FIG. **1** is a perspective view of an on-board ammonia dissociation system **100** for an internal combustion engine. The on-board ammonia dissociation system **100** is described in commonly owned U.S. Pat. No. 11,981,562, issued on May 14, 2024, entitled "SYSTEMS AND METHODS FOR THE ON-BOARD CATALYTIC PRODUCTION OF HYDROGEN FROM AMMONIA USING A HEAT EXCHANGE CATALYST UNIT AND AN ELECTRIC CATALYST UNIT OPERATING IN SERIES", which is incorporated by reference herein.
(25) In an embodiment, an ammonia liquid tank **102** is mounted to a motor vehicle or engine. The ammonia liquid tank **102** can be coupled to a pump. In an embodiment, the tank **102** is refillable and/or replaceable.
(26) In an embodiment, a temperature control valve **104** receives a temperature feedback signal that contains a temperature reading from an electric catalyst unit **106** during a cold start of the engine. The temperature feedback signal can be generated by a temperature sensor coupled to the electric catalyst unit **106**. Once the electric catalyst unit **106** reaches a threshold temperature (i.e., the temperature reading is equal to or greater than the threshold temperature) suitable to perform the ammonia dissociation process, the temperature control valve **104** opens and the gaseous ammonia passes through the heat exchange catalyst unit **108**, and travels downstream to the electric catalyst unit **106**, which is heated using power supplied from the power supply.
(27) In an embodiment, a controller (not shown) can be coupled to components of the on-board ammonia dissociation system **100** to receive inputs from various sensors, such as temperature sensors and pressure transducers, and control, for example, operation of the temperature control valve **104**, the pressure control valve **110**, and heating elements of the electric catalyst unit **106**.
(28) In another embodiment, the controller can be integrated into the hardware and software with the vehicle's electronic control unit (ECU). In this embodiment, the flow rate of hydrogen fed to the injection system of the engine can be measured and reported back to the ECU as a mechanism to control the injection strategy.
(29) If the electric catalyst unit **106** has not reached the threshold temperature, then the temperature control valve **104** continues to monitor the temperature feedback signal and prevents the downstream travel of the gaseous ammonia to the electric catalyst unit **106**.
(30) In an embodiment, the temperature of the heated exhaust gas entering the heat exchange catalyst unit **108** is judged based on the current draw in the electric catalyst unit **106**, where the current draw is indicative of how effective the heat exchange catalyst unit **108** is in cracking the gaseous ammonia.
(31) For example, if there is hydrogen and nitrogen passing from the heat exchange catalyst unit **108** to the electric catalyst unit **106**, the heating elements of the electric catalyst unit **106** will draw minimal or no current as the ammonia dissociation process will not be required (except for any residual ammonia flowing with the hydrogen and nitrogen).
(32) Conversely, if gaseous ammonia passes from the heat exchange catalyst unit **108** to the electric catalyst unit **106**, the heating elements of the electric catalyst unit **106** will draw current and energize to generate heat for cracking.
(33) However, during a normal or high load operating conditions of the engine (i.e., not during a cold start or low load operating conditions), the on-board ammonia dissociation system **100** does not utilize the electric catalyst unit **106** to perform the ammonia dissociation process, and the heat exchange catalyst unit **108** performs the ammonia dissociation process as it will have been heated to the threshold temperature by the exhaust gas from the engine.
(34) In an embodiment, the pressure control valve **110** is located in series with the temperature control valve **104**, and controls the amount of gaseous ammonia which is fed into the heat exchange catalyst unit **108**.
(35) In an embodiment, to facilitate a cold start of the on-board ammonia dissociation system **100**

when the exhaust gas from the engine is not at a threshold temperature suitable to perform the ammonia dissociation process, the electric catalyst unit **106** is used to heat the catalyst so that the gaseous ammonia can be cracked, and the resulting hydrogen is to be supplied to the downstream injection system for the engine as a co-fuel along with ammonia. The engine can then burn the hydrogen and ammonia as co-fuels, powering the engine which results in heated exhaust gas being supplied to the on-board ammonia dissociation system **100**.

(36) FIG. **2** is a perspective view of an electric catalyst unit **106**, according to an embodiment of the present invention. The electric catalyst unit **106** comprises a housing **200**, an inlet **202**, an outlet **204**, and covers **206**, **208** coupled to opposing ends of the housing **200**.

(37) In an embodiment, the housing **200** includes multiple sections **210**, such as the four sections **210***a-d* depicted in FIG. **2** which are coupled together. The four sections **210***a-d* depicted in FIG. **2** are intended to be exemplary and are not limiting, and the housing **200** can include any number of sections **210**, from a minimum of two sections up to any number of sections that may be required based on a specific engine size, footprint, and/or heating requirements.

(38) In an embodiment, each of the sections **210** have the same dimensions, such as the same diameter and the same width. In another embodiment, the sections can have variable dimensions, such that sections of different widths can be coupled together to form the housing **200**. In yet another embodiment, the sections can have different diameters, such that the annular space within the housing varies based on the diameter of each section. For example, sections having varying diameters can be utilized to form a variety of housing shapes, such as a circular cone, a dual-cone shape, an inverted dual-cone shape, and other polyhedron shapes where the diameter is not constant along the length.

(39) In an embodiment, the sections **210** form a housing **200** with a circular cylinder shape. In other embodiments, the sections **210** can form housings having the shape of an elliptical cylinder, a polygonal cylinder, a cube, a cuboid, a triangular prism, a rectangular prism, and the like.

(40) In an embodiment, the sections **210** are coupled together at joints **212**. The joints **212** can be formed via welding, soldering, riveting, bolting, brazing, and/or by using mechanical fasteners to press fit opposing sections **210** together. In an embodiment, the joints **212** can include a gasket, such as a beaded gasket, to improve sealing between each of the opposing surfaces the sections **210**.

(41) In an embodiment, the sections **210** are removably coupled to each other, such that different sections can be used to form a modular housing **200**. The ability to remove sections **210** from one another allows for servicing, repair and/or replacement of sections **210** and/or housing **200** as needed.

(42) In an embodiment, each section **210** is formed from a nickel alloy, such as Inconel®.

(43) In an embodiment, the electric catalyst unit **106** includes covers **206**, **208** disposed on opposite ends of the housing **200**. The covers **206**, **208** are removably coupled to the housing **200** such that they can be removed to service or replace the components within the housing **200**. In another embodiment, only one of the covers **206**, **208** is removable. In an embodiment, the inner surfaces of the covers **206**, **208** can be coated with a ceramic paste that forms a thermal barrier and increases the thermal efficiency of the electrical catalyst unit **106**. In yet another embodiment, a gasket, such as a beaded gasket, is placed between the covers **206**, **208** and the housing **200** to improve sealing between the two opposing components.

(44) In an embodiment, the inlet **202** is disposed on cover **206**, and the outlet **204** is disposed on cover **208**. The inlet **202** and outlet **204** can be removably attached to respective covers **208**, **208** so that different inlets and outlets having various dimensions, sizes, and flow properties can be utilized with the electric catalyst unit **106** in a modular fashion.

(45) In an embodiment, the inlet **202**, the outlet **204**, and the covers **206**, **208** can be made from the same metallic material as the sections **210**. In another embodiment, the inlet **202**, the outlet **204**, and the covers **206**, **208** can be made from stainless steel, silver, bronze, and comparable alloys. In

an embodiment, the covers **206**, **208** seal the electric catalyst unit **106** in an air-tight fashion.
(46) Each section **210** of the electric catalyst unit **106** includes at least one respective power feed-through **214** for supplying electric current to heating elements disposed within housing **200**, and at least one respective temperature sensor **216**. The power feed-through **214** is described in commonly owned U.S. Pat. No. 12,009,650, issued on Jun. 11, 2024, entitled "APPARATUS FOR AN ELECTRIC FEEDTHROUGH FOR HIGH TEMPERATURE, HIGH PRESSURE, AND HIGHLY CORROSIVE ENVIRONMENTS", which is incorporated by reference herein.
(47) Each power feed-through **214** is coupled on one end to a power supply, which can be a traditional 12V vehicle battery, or can be a dedicated power supply that is separate from the vehicle battery. The power supply provides electrical current to heating elements disposed within the housing **200**, as described in more detail herein with respect to FIG. **3**.
(48) In an embodiment, the dedicated power supply does not provide power to any other vehicle electrical system or to the internal combustion engine, and the power supply only supplies power to the electric catalyst unit **106**.
(49) In an embodiment, the power supply consists of one or more 8V, 12V, 24V or 48V batteries which are connected in series or parallel. The multiple batteries share the load demand from each heating element disposed within the electric catalyst unit **106**.
(50) In yet another embodiment, the power supply can be a power pack having multiple lithium-ion (Li-ion) batteries, nickel-metal-hydride (NiMH) batteries, lead-acid batteries, ultracapacitors, and the like.
(51) In an embodiment, a battery management controller balances and optimizes the power output from each battery in the power supply. The battery management controller protects the power supply, and each battery contained therein, from various fail mode conditions such as overcurrent protection, over/under voltage conditions, and over/under temperature conditions.
(52) The battery management controller can also measure and estimate states and parameters of each battery such as state of charge, state of power, state of health, internal resistance, usable capacity, operating temperature, estimated duty cycle, and the like.
(53) In another embodiment, the battery management controller, or its functions, can be incorporated into the vehicle ECU.
(54) FIG. **3** is a cross-sectional side view of the electric catalyst unit **106**, according to an embodiment of the present invention. In an embodiment, each section **210** of the housing **200** contains a respective heating element **300** that is coupled to a respective power feed-through **214** (which is a positive terminal) and to a respective ground/negative terminal **218**. Electric current from the power supply is configured to flow from each power feed-through **214** to its respective heating element **300**.
(55) In an embodiment, each section **210** includes at least one temperature sensor **216** that is attached via a radial fitting on the section **210**. The temperature sensor **216** can be a thermocouple, a thermistor, a resistance temperature detector (RTD), a semiconductor-based sensor, an analog thermometer integrated circuit, or a digital thermometer integrated circuit. In a preferred embodiment, the temperature sensor **216** is a thermocouple.
(56) Since each section **210** includes a respective heating element **300** and a respective temperature sensor **216**, the output gas temperature within each section **210** can be monitored, and the gas can be continuously heated, as needed, as it flows through each successive section **210** within the electric catalyst unit **106**.
(57) Arrow **302** indicates the direction of gas flow through the electric catalyst unit **106**, where gas enters at the inlet **202** and flows through each section **210***a-d* toward the outlet **204**. On each section **210**, the temperature sensor **216** is positioned at a location downstream from the heating element **300**. This configuration allows the output gas temperature at each section **210** to be detected.
(58) For example, as gas flows from section **210***a* to section **210***b*, the temperature sensor **216***a* in

section **210***a* detects the temperature of the air, gas, and/or the air-gas mixture (collectively referred to herein as “gas” or “gas flow”) flowing through section **210***a*. Based on a comparison of the output gas temperature with a threshold temperature, the heater percentage of the heating element **300***a* within section **210***a* is adjusted, as described in more detail herein.
(59) The threshold temperature can range from 400° C. to 700° C., and in a preferred embodiment, the threshold temperature is at least 600° C. When the output gas temperature of a section is greater than or equal to the threshold temperature, this indicates that the gas flow contains little or no ammonia and contains primarily hydrogen and nitrogen—the gaseous ammonia has successfully been cracked within the electric catalyst unit **106** or the gaseous ammonia has previously been cracked in the heat-exchange catalyst unit **108** prior to flowing to the electric catalyst unit **106**.
(60) In this scenario, the heater percentage of the heating element is not increased, and the downstream heating element(s) within the electric catalyst unit **106** are not energized. However, if the output gas temperature of a section is less than the threshold temperature, then the heater percentage of the heating element is increased based on the temperature difference and gas flow rate, as described in more detail herein with respect to FIG. **9**. In this manner, excessive or unneeded electric current is not drawn from the power supply, and only the power required to maintain the output gas temperature at the threshold temperature throughout each section **210** of the electric catalyst unit **106** is supplied. This sequential energization of each heating element **300** within the electric catalyst unit **106** preserves the charge in the power supply and reduces unnecessary power draws, which helps to maintain the health and lifespan of the power supply.
(61) FIG. **4** is a cross-sectional perspective view of the electric catalyst unit, and FIG. **5** is a lateral side view of the electric catalyst unit **106**, according to an embodiment of the present invention. In an embodiment, each section **210** is mounted at a different rotational angle (or offset) relative to its adjacent section. This mounting results in each of the power feed-throughs **214** and temperature sensors **216** being offset relative to one another, as shown in FIG. **5**.
(62) In addition, this mounting results in each heating element **300***a-d* to be positioned with a different rotational orientation relative its adjacent heating elements. This offset arrangement of the heating elements **300***a-d* results in a turbulent path for the gas as it traverses through the sections **210** of the electric catalyst unit **106**. Gas flowing from the inlet **202** to the outlet **204** of the electric catalyst unit **106** cannot not flow in a straight linear path due to the overlapping nature of the portions of the heating elements **300**; there are no straight linear paths or channels that provide an unobstructed flow of gas between the inlet **202** and outlet **204** of the electric catalyst unit **106**. Instead, as gas contacts and/or collides with the heating elements **300**, the gas is forced to travel around the heating elements **300**, causing randomized and non-linear gas paths throughout the electric catalyst unit **106**.
(63) In an embodiment, each section **210** is mounted with a rotational offset of 5 degrees to 45 degrees relative to its adjacent section. In an embodiment, the degree of rotational offset between each adjacent section **210** is equal. In another embodiment, the degree of rotational offset between each adjacent section **210** is varied.
(64) By mounting each section **210** with a rotational offset, the location of the power feed-throughs **214** and the temperature sensors **216** is varied along the length of the housing **200**. This staggered placement of the power feed-throughs **214** and temperature sensors **216** allows the respective heating elements **300** within each section **210** to be energized at different angular locations within the inner circumference of the housing **200**, and similarly, allows for temperature detection at different angular locations within the inner circumference of the housing **200**.
(65) FIG. **6** is a perspective view of a section **210** of the housing **200** of the electric catalyst unit **106**, according to an embodiment of the present invention. The section **210** includes the heating element **300** coupled to the power feed-through **214** which is a positive terminal and coupled on its opposite end to the ground/negative terminal **218**. In an embodiment, the heating element **300** is a wire heater with a serpentine path as shown in FIG. **5**. The heating element **300** is not limited to the

wire heater depicted in FIG. **5** and can be in the form of an air process heater, a cartridge heater, a tubular heater, a band heater, a strip heater, a spiral heater, a ribbon wire heater, an etched foil heater (or a thin-film heater), a ceramic heater, a ceramic fiber heater, a lattice heating element, and the like.

(66) In another embodiment, the opposite end of the heating element **300** can be directly connected to the section **210** (i.e., the housing **200**) which serves as a negative terminal or a ground terminal.

(67) In an embodiment, the heating element **300** is removably secured to the section **210**, such that various types, forms, and shaped heating elements can be interchangeably utilized with the electric catalyst unit **106** in a modular fashion. In yet another embodiment, each section **210** of the housing **200** can utilize a different type of heating element.

(68) In another embodiment, the heating element **300** is a three-dimensionally printed lattice heating element as described in commonly owned U.S. patent application Ser. No. 18/827,074, filed on Sep. 6, 2024, entitled “SYSTEM AND METHOD FOR A THREE-DIMENSIONALLY PRINTED LATTICE STRUCTURE FOR HEATING GAS IN A NON-LINEAR PATH”, which is incorporated by reference herein.

(69) In an embodiment, the section **210** can include at least one additional radial fitting that can receive the temperature sensor **214** (not shown in FIG. **6**). The additional radial fitting may be used for a variety of functions. For example, the additional radial fitting may serve as an inlet, outlet, or may be coupled to equipment for temperature, throughput, and/or pressure sensing, such as additional temperature sensors, transducers, flow meters, and the like.

(70) In an embodiment, the surfaces of the heating element **300** are coated with a catalyst that facilitates the ammonia dissociation process. The catalyst can be coated to the heating element **300** using a washcoating or deposition technique to bind or adhere the catalyst to the surfaces of the heating element **300**. In an embodiment, the inner surface **600** of the section **210** can also be coated with the catalyst. The catalyst can be coated to the inner surface **600** using a washcoating or deposition technique to bind or adhere the catalyst to the inner surface **600**.

(71) In an embodiment, the inner surface **600** is coated with a ceramic, causing the section **210** to act as an insulator and allowing heat to be focused and reflected toward the heating element **300**, which in turn promotes heating of the catalyst.

(72) FIG. **7** is an inlet end view of the electric catalyst unit **106**, according to an embodiment of the present invention. FIG. **7** depicts an exemplary rotational offset of 30 degrees between each section **210**, such that there is a variance of 90 degrees between the mounting of section **210***a* relative to the rotational mounting of section **210***d*.

(73) FIG. **8** is an inlet end view of the electric catalyst unit **106** of FIG. **6** with the cover **206** removed, according to an embodiment of the present invention. As shown, the rotational offset mounting of each section **210***a-d* results in each heating element **300***a-d* to be positioned with a different rotational orientation relative its adjacent heating elements, preventing a straight linear path or channel for gas flow through the electric catalyst unit **106**.

(74) In an embodiment, catalyst, such as discrete catalyst media or powder, is deposited into the housing **200** around the heating elements **300**.

(75) FIG. **9** is a flowchart depicting sequential energization of heating elements **300** as gas flows through the electric catalyst unit **106**, according to an embodiment of the present invention. In an embodiment, the controller utilizes a temperature feedback signal from the temperature sensor **216** of each section **210** to control the power supplied to the respective heating elements **300**. The controller sequentially energizes each heating element **300** as needed to maintain the output gas temperature at the threshold temperature throughout the electric catalyst unit **106**. By maintaining the threshold temperature throughout the electric catalyst unit **106**, the endothermic reaction required for ammonia dissociation can occur continuously as the gas flow traverses through each section **210**.

(76) The threshold temperature can range from 400° C. to 700° C., and in a preferred embodiment,

the threshold temperature is at least 600° C. When the output gas temperature of a section **210** is greater than or equal to the threshold temperature, this indicates that the gas flow contains little or no ammonia and contains primarily hydrogen and nitrogen (i.e., the gaseous ammonia has been cracked).

(77) At step **900**, gas enters the electric catalyst unit **106** via the inlet **202**. If the ammonia dissociation process was successfully performed in the heat exchange catalyst unit **108**, then the resulting gas flow of hydrogen and nitrogen gas (as well as residual ammonia gas) will flow to the electric catalyst unit **106**. In this scenario, there will be little or no ammonia in the gas flow that causes cooling within the sections **210** of the electric catalyst unit **106**. Thus, some or all the heating elements **214** may not be required to be energized at all, energized fully, or energized for a full heating cycle. For example, if pre-heated residual ammonia needs to be cracked in the electric catalyst unit **106**, then only the first heating element could be partially energized, instead of being fully energized, and the remaining downstream heating elements could remain off (i.e., not energized and not drawing current).

(78) If, however, the ammonia dissociation process was not performed in the heat exchange catalyst unit **108**, then gaseous ammonia will flow to the electric catalyst unit **106**. The gaseous ammonia may be pre-heated by the heat exchange catalyst unit **108** if the engine was operating at a low load operating condition, or the gaseous ammonia could be cold and not pre-heated if the engine was operating in a cold start condition.

(79) Referencing the electric catalyst unit **106** depicted in FIG. **3** having four sections **210***a-d*, at step **902**, the output gas temperature is detected by the temperature sensor **216***a*. At step **904**, the controller receives a temperature feedback signal indicating the output gas temperature of section **210** and compares the output gas temperature with the threshold temperature.

(80) If the output gas temperature of section **210***a* is greater than or equal to the threshold temperature, the process continues to step **906** and the controller does not adjust the heater percentage of heating element **300***a*, and does not energize the remaining downstream heating elements **300***b-d*. The gas flow continues through the remaining sections of the electric catalyst unit **106** (if any) and exits via the outlet **204** at step **908**. At this step, the gas flow comprises a gas mixture of resulting hydrogen and nitrogen components, and is supplied as fuel, or co-fuel along with ammonia, to an injection system for the internal combustion engine.

(81) If, however, the output gas temperature of section **210***a* is less than the threshold temperature at step **904**, then the process continues to step **910** where the controller determines if the heater percentage of heating element **300***a* is at its maximum value (i.e., fully energized and drawing its maximum current from the power supply). If the heating element **300***a* is not at its maximum heater percentage, then the process continues to step **912** where the controller increases the heater percentage of heating element **300***a* (i.e., supplies a higher current/more power).

(82) In an embodiment, the controller increases the heater percentage of heating element **300***a* (i.e., the amount of current drawn by the heating element) proportionally based on the magnitude of the difference between the output gas temperature and the threshold temperature, as well as the gas flow rate through the section **210***a*.

(83) For example, during a cold start or low load operating condition of the engine, the gas flow rate is relatively low, and the heating element **300***a* may be able to heat section **210***a* to the threshold temperature so that ammonia cracking can occur within section **210***a*. As the gas flow rate increases, heating element **300***a* may not generate sufficient heat to fully crack all the gas entering the electric catalyst unit **106**. This results in uncracked gaseous ammonia travelling downstream to the next section **210***b* of the electric catalyst unit **106**. In this scenario, heating element **300***a* serves as a pre-heater for the adjacent downstream heating element **300***b*.

(84) However, if the engine is operating at a normal or high load operating condition, then the gaseous ammonia is cracked in the heat exchange catalyst unit **108**, and residual ammonia flows from the heat exchange catalyst unit **108** to the electric catalyst unit **106**. This residual ammonia

has been pre-heated by the heat exchange catalyst unit **108**, thus the heating element **300***a* may not need to be energized to its full heater percentage to maintain the output gas temperature of section **210***a* at the threshold temperature.
(85) In an embodiment, if the difference between the output gas temperature and the threshold temperature is relatively large, the controller proportionally increases the heater percentage of heating element **300***a* by higher amount. This results in a higher current draw from the power supply by the heating element **300***a*.
(86) However, if the temperature difference is relatively small, the controller proportionally increases the heater percentage of heating element **300***a* by a smaller amount. This results in a relatively lower current draw from the power supply by the heating element **300***a*.
(87) Once the heater percentage of the heating element **300***a* is increased, the process returns to step **902** where the output gas temperature of section **210***a* is detected by the temperature sensor **216***a*.
(88) If, however, at step **910** the heater percentage of heating element **300***a* is at its maximum value, the process continues to step **914** where the adjacent downstream heating element **300***b* is energized. In an embodiment, the heater percentage of the downstream heating element **300***b* is controlled based on the temperature difference and gas flow rate. The process returns to step **902** where the output gas temperature of section **210***b* is detected by the temperature sensor **216***b*.
(89) By increasing the heater percentage of heating element **300***a* to its maximum value as needed based on the output gas temperature, and only sequentially energizing the downstream heating element **300***b* if heating element **300***a* cannot generate sufficient heat while operating at its maximum heater percentage, excessive or unneeded electric current is not drawn from the power supply. Thus, the controller (a) does not supply power to a downstream heating element if an upstream heating element generates sufficient heat for cracking, and (b) only applies the heater percentage required for the upstream heating element to generate sufficient heat for cracking. Over time, this mitigates degradation and discharge of the power supply and maintains the health and life of the power supply.
(90) In an embodiment, the controller utilizes an algorithm that factors the temperature difference between the output gas temperature and the threshold temperature, as well as the gas flow rate, to determine the heater percentage of heating element **300**. In an embodiment, the temperature difference and heater percentage applied to heating element **300** have an inverse relationship.
(91) In yet another embodiment, the heater percentage can be adjusted incrementally by a set amount for each heating iteration. For example, if the heater percentage of the heating element **300** is not at its maximum value, the controller can continue to increment the heater percentage by the set amount until the maximum value is reached or until the output gas temperature of that respective section **210** has reached the threshold temperature. The set amount can be any increment, such as 0.5% up to 25%, for example.
(92) In another embodiment, a heating cycle time of the heating element **300** is determined by the controller based on the magnitude of the temperature difference and/or the gas flow rate. For example, if the temperature difference is relatively large, then the heating element can be proportionally energized for a larger period of time. Conversely, if the temperature difference is relatively small, then the heating element can be proportionally energized for a smaller period of time (i.e., a period less than a full heating cycle), resulting in a smaller duration of current draw by the heating element **300** from the power supply.
(93) In an embodiment, the controller can adjust both the heater percentage of the heating element **300** as well as the heating cycle time. By optimizing the percentage and duration of the current draw to be as small and short as possible, while still achieving the desired threshold temperature, unnecessary power draws are prevented, which helps to maintain the health and lifespan of the power supply.
(94) In another embodiment, instead of a separate power feed-through coupled to each section **210**,

a switching mechanism integrated with, or coupled to, the electric catalyst unit **106** can be utilized to provide current to multiple heating elements. For example, a single power feed-through can supply power to a first heating element in a first section. A switching device, such as an electric relay, can be coupled between the first heating element and a second heating element in an adjacent second section. If the controller determines that power should be supplied to the second heating element, the electric relay can close, creating an electric circuit between the two heating elements, thereby allowing current to be drawn by the second heating element.
(95) FIG. **10** is a flowchart depicting sequential energization of heating elements as gas flows through the electric catalyst unit **106**, according to an embodiment of the present invention. FIG. **10** depicts a similar process as described with reference to FIG. **9**. However, in FIG. **10**, at step **1006** wherein when the heater percentage of heating element **300***a* is not adjusted, the process continues to step **1008**, where the controller determines if there is a downstream section **210** in the housing **200** of the electric catalyst unit **106**.
(96) If at step **1008** the controller determines there is not a downstream section **210** in the housing **200**, the gas flow exits the electric catalyst unit **106** via the outlet **204** at step **1010**. At this step, the gas flow comprises a gas mixture of resulting hydrogen and nitrogen components, and is supplied as fuel, or co-fuel along with ammonia, to an injection system for the internal combustion engine.
(97) However, if at step **1008** the controller determines there is a downstream section **210** in the housing **200**, the process returns to step **1002** where the output gas temperature of the adjacent downstream section **210** is detected, and the heater percentage of heating element **300***a* is increased, or the downstream heating element **300***b* is sequentially energized if the heating element **300***a* is already operating at its maximum heater percentage.
(98) In this manner, the controller can continuously monitor the output gas temperature at each section **210** and sequentially energize heating elements **300** within the electric catalyst unit **106** as needed. For example, in a scenario where the output gas temperature suddenly drops, such as when the engine goes from a normal operating load to an low load operating condition (i.e., such as at idle during a stop light or in stop-and-go traffic), the controller can sequentially energize additional heating elements or increase the heater percentage of one or more heating elements so that the ammonia cracking process can continue without interruption.
(99) Conversely, in a scenario where the output gas temperature suddenly increases, such as when the engine goes from a low load operating condition to a normal or high load operating condition (i.e., such as sudden acceleration after stopping at a stop light or intersection), the controller can proportionally or incrementally reduce the heater percentage of heating elements, or stop supplying power to heating elements, so that excessive or unneeded electric current is not drawn from the power supply.
(100) FIG. **11** is a block diagram of the control system for the electric catalyst unit **106**, according to an embodiment of the present invention. The controller **1100** receives a temperature feedback signal from a temperature sensor **216** and compares the output gas temperature to the threshold temperature, as described herein. Based on this comparison, the controller **1100** controls the heater percentage applied to the heating element **300**.
(101) In an embodiment, the controller can utilize an artificial intelligence engine **1102** to collect data over time to more efficiently and quickly control the heating elements **300**. For example, in addition to output gas temperatures and temperature differences, data such as ambient temperature, engine temperature, time of day, weather conditions, road congestion and traffic, navigation and route data, driving behavior, and the like can be stored in a learning database **1104**.
(102) A large language model (LLM) **1106** can analyze the data in the learning database **1004** to anticipate or predict current and/or future power supply requirements based on historical patterns. For example, if a driver of the vehicle travels a route daily that has multiple stop signs or stop lights, or which is routinely heavily congested (i.e., stop-and-go traffic), the LLM **1106** can determine that the electric catalyst unit **106** will likely be utilized during low load operating

conditions of the engine during stops, idling, or slow movement. The controller **1100** can proactively energize heating elements based on the analysis of the LLM **1106**.
(103) Thus, by anticipating the power supply requirements for the heating elements **300**, the controller **1100** can quickly energize the heating elements based on feedback from the artificial intelligence engine **1002** without requiring the controller to fully perform its temperature difference and power supply determination as described herein.
(104) In an embodiment, the artificial intelligence engine **1102** is integrated with, or coupled to the controller **1100**. In another embodiment, the artificial intelligence engine **1102** is located remotely, such as on cloud-based LLM or other cloud-based computing device or server.
(105) While the principles of the disclosure have been illustrated in relation to the exemplary embodiments shown herein, the principles of the present invention are not limited thereto and include any modification, variation, or permutation thereof.

Claims

1. An electric catalyst unit for ammonia dissociation, comprising: a housing comprising a first section and a second section, the first section containing a first heating element and the second section containing a second heating element; a first power feed-through coupled to the first heating element and to a power supply; a second power feed-through coupled to the second heating element and to the power supply; a first temperature sensor coupled to the first section; a second temperature second coupled to the second section; and a controller communicatively coupled to the power supply, the first temperature sensor, and the second temperature sensor, wherein the controller increases a heater percentage of the first heating element if a temperature detected by the first temperature sensor is below a threshold temperature, and wherein the controller supplies power to the second heating element only if (i) the first heating element is operating at a maximum heater percentage and (ii) the temperature detected by the first temperature sensor is below the threshold temperature.
2. The electric catalyst unit of claim 1, wherein the threshold temperature is a temperature sufficient to perform ammonia dissociation.
3. The electric catalyst unit of claim 1, wherein the threshold temperature is at least 600° C.
4. The electric catalyst unit of claim 1, where the first heating element and the second heating element are selected from a group consisting of an air process heater, a cartridge heater, a tubular heater, a band heater, a strip heater, a spiral heater, a ribbon wire heater, an etched foil heater, a ceramic heater, a ceramic fiber heater, and a lattice heating element.
5. The electric catalyst unit of claim 1, wherein the first heating element and the second heating element are coated with a catalyst that facilitates ammonia dissociation.
6. The electric catalyst unit of claim 1, wherein the first heating element and the second heating element are mounted within the housing at a rotational offset relative to one another.
7. The electric catalyst unit of claim 1, wherein the first temperature sensor and the second temperature sensor are mounted on the housing at a rotational offset relative to one another.
8. An electric catalyst unit for ammonia dissociation, comprising: a housing comprising a first section and a second section, the first section containing a first heating element and the second section containing a second heating element; a first power feed-through coupled to the first heating element and to a power supply; a second power feed-through coupled to the second heating element and to the power supply; a first temperature sensor mounted to the first section downstream from the first heating element; a second temperature second mounted to the second section downstream from the second heating element; and a controller communicatively coupled to the power supply, the first temperature sensor, and the second temperature sensor, wherein the controller increases a heater percentage of the first heating element if a temperature detected by the first temperature sensor is below a threshold temperature, and wherein the controller supplies

power to the second heating element only if (i) the first heating element is operating at a maximum heater percentage and (ii) the temperature detected by the first temperature sensor is below the threshold temperature.
9. The electric catalyst unit of claim 8, wherein the threshold temperature is a temperature sufficient to perform ammonia dissociation.
10. The electric catalyst unit of claim 8, wherein the threshold temperature is at least 600° C.
11. The electric catalyst unit of claim 8, where the housing is hermetically sealed.
12. The electric catalyst unit of claim 8, wherein the power supply is a vehicle battery.
13. The electric catalyst unit of claim 8, wherein the first heating element and the second heating element are mounted within the housing at a rotational offset relative to one another.
14. The electric catalyst unit of claim 8, wherein the first temperature sensor and the second temperature sensor are mounted on the housing at a rotational offset relative to one another.
15. An electric catalyst unit for ammonia dissociation, comprising: a housing comprising a first section and a second section, a first heating element disposed within the first section, the first heating element having a positive end coupled to a first power feed-through and having a negative end coupled to a first ground terminal; a second heating element disposed within the second section, the second heating element having a positive end coupled to a second power feed-through and having a negative end coupled to a second ground terminal; a first temperature sensor mounted to the first section at a location downstream from the first heating element; a second temperature second mounted to the second section at a location downstream from the second heating element; and a controller communicatively coupled to a power supply, the first temperature sensor, and the second temperature sensor, wherein the controller increases a heater percentage of the first heating element if a temperature detected by the first temperature sensor is below a threshold temperature, and wherein the controller supplies power to the second heating element only if (i) the first heating element is operating at a maximum heater percentage and (ii) the temperature detected by the first temperature sensor is below the threshold temperature.
16. The electric catalyst unit of claim 15, wherein the threshold temperature is a temperature sufficient to perform ammonia dissociation.
17. The electric catalyst unit of claim 15, wherein the threshold temperature is at least 600° C.
18. The electric catalyst unit of claim 15, where the temperature detected by the first temperature sensor provides an indication of whether ammonia dissociation is occurring as gas flows through the first section.
19. The electric catalyst unit of claim 15, wherein the first heating element and the second heating element are coated with a catalyst that facilitates ammonia dissociation.
20. The electric catalyst unit of claim 15, wherein the first heating element and the second heating element are mounted within the housing at a rotational offset relative to one another.